

# LED CAVE - New Dimensions for Large-scale Immersive Installations

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## ABSTRACT

Virtual reality is an established tool in many areas of application, including scientific visualisation. Hardware such as head-mounted displays and immersive installations can be utilised to analyse and display data. For the most part, these types of installations are based on projectors, making them complex and require a large space.

We present the concept of a new CAVE system based on the latest generation of LED panels. We report our preliminary findings based on 6 months of operation. This paper focuses on the implementation, the comparison to a traditional projector-based CAVE and early user feedback. We discuss the potential future for large-scale immersive installation systems and the possibilities opened up by the advances in LED technology.

**Keywords:** Virtual Reality, Large-scale Displays, Scientific Visualization, LED, Stereoscopic 3D, CAVE.

## 1 INTRODUCTION

Virtual Reality (VR) has become a widely applied medium within a multitude of usage scenarios ranging from gaming and entertainment in home setups to industrial applications and scientific visualisations [2, 13, 15, 17]. Apart from various compact head-mounted displays (HMDs), VR environments can also be provided by large-scale installations, one example being CAVE-like systems, the concept of which was originally suggested in 1992 [5].

Often, CAVE systems rely on projectors to display images onto their three to six walls working as screens [20]. Due to the space requirements of the projectors, CAVEs are typically large-scale complex installations that may even span multiple floors.

In this paper, we present the concept of a 5-sided CAVE system based entirely on LED display technology. As of writing this paper, we are not aware of another publication or CAVE system using LED technology for a 5- or 6-sided CAVE featuring stereoscopic displays and user tracking. The LED panels create screens without visible borders or bezels in contrast to LCD panels, akin to CAVE systems using back projection. However, as the walls themselves are the screens, the space requirements for the entire system are barely larger than the interior space of the CAVE. We report our early findings from the first few months of operating the LED-based CAVE and compare them to the experiences with our previous, projector-based CAVE system. In this paper, we discuss the basic construction, technical specifications and early user feedback as well as provide an outlook on the potential future of large-scale immersive VR installations.

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## 2 BACKGROUND OF LARGE-SCALE INSTALLATIONS

VR refers to an immersive, interactive, multi-sensory, viewer-centered, 3D computer-generated environment [21]. It has found application in many areas, including entertainment, art history [13], education [17], medicine [3], and research and scientific visualisation [2]. Today, VR applications are commonly experienced through HMD-style displays. However, large-scale immersive installations [1] are also used in many research fields [6, 7, 19, 24, 25, 27] ranging from data analysis and visualisations to education and science communication.

Typically, large-scale immersive installations take the form of CAVEs or CAVE-like systems or Powerwalls. Many use projection and offer users immersive experiences with at least one user being tracked. There are many forms of CAVEs [20]. Muhanna [20] defines the difference between them as follows: CAVEs are fully immersive room-based systems and powerwalls encompass non-room-based and partially immersive installations. As powerwalls are not the main focus of this paper, we will focus on CAVE systems. The first CAVE was built and designed in 1991 [5]: it is a room-scale cube that uses rear projection onto three side walls to display a virtual world and a top projection for the floor. Nowadays, a CAVE or CAVE-like system is characterised by a room-scale area (not necessarily cubic) with 3-6 walls acting as screens and user tracking. Often, projectors are used with a majority of modern CAVEs using back-projection to generate an immersive experience [9]. A notable feature of many CAVE systems, including the original CAVE from 1991, is the ability to collaborate in person as they can accommodate small groups simultaneously.

Settgast et al. found that a CAVE system gives users more freedom regarding body perception and small-scale movement than HMDs [23]. Their results also indicate a slightly higher feeling of immersion in the CAVE system in addition to a higher feeling of cybersickness. Cordeil et al. [4] also compared an HMD with a CAVE system in the context of collaborations with a user study and found participants felt it easier to communicate when collaborating in a CAVE system.

There are in parallel debates on whether CAVE systems are still relevant [8] given the challenges, including costs and maintenance issues. In their review of 25 years of maintaining a visualisation center with immersive installations, Marai et al. [16] acknowledge these challenges but also highlight advantages of large-scale immersive displays. These include group work and collaboration as well as immersion in the projects.

### 2.1 Projection-based CAVEs

Traditionally, CAVEs have been projection-based with rear-projection to avoid occlusion and shadows by users on the walls [9]. There have been developments to the CAVE system concept, such as the StarCAVE [10], which was comprised of five walls with floor projection using 15 rear-projected wall displays. In comparison, the DAVE system (Definitely Affordable Virtual Environment) [14] consists of four walls with a combination of rear- and front-projection. A more high-end version of the CAVE may be found in the Cornea of the King Abdullah University of Science and Technology [9] with 6 walls with rear-projection and reported to be the brightest CAVE

in 2010. A larger CAVE system would be the WorldViz CAVE [7], which stands at  $5.2\text{ m} \times 5.2\text{ m} \times 2.7\text{ m}$  with four walls and uses eight projectors for front projection. The aixCAVE is just as large as the WorldViz [11] but offers back projection and higher quality (approximately 4k per wall) with 24 projectors. Further noteworthy systems include the AquaCAVE, which replaced the five sides of a swimming pool with rear-projection screens [28]. As of writing this paper, contemporary CAVE systems usually have four walls and are rear-projected.<sup>1,2,3</sup>

## 2.2 Non-projection-based CAVEs

Besides the projection-based CAVEs, there are other systems that utilise different technologies. The MobiCAVE is a simplified version of a CAVE system [26]. It has three walls consisting of LCD screens and floor projections but no stereo glasses however it could support 3D projection. Advantages include modular and transportable setup and multiple users. A major disadvantage is the bezels of the LCD screens that are clearly visible. Another system based on LCDs is the CAVE2 [12], which also has visible bezels, however, this setup can be done in cylindrical or curved forms but doesn't include a ceiling. The NexCAVE is based on 3D-TVs, which offers more brightness in contrast to projectors [22], however, the borders of the TVs are still visible. Both systems use LCDs to create large walls to surround the users. A major issue with non-projection-based CAVEs (or panel-based CAVEs) is the floor, as it needs to carry the weight of users next to being a display [9]. Additionally, LCD screens and panel-based CAVEs have the issue of screen borders or bezels.

## 3 CONCEPT OF THE LED CAVE

For over ten years we have been operating projector-based large-scale VR installations in the shape of a Powerwall and a CAVE [18, 19]. In 2018, we acquired our first LED-based installation: a Powerwall capable of displaying stereoscopic content. This technology builds a screen from a number of LED panels. As these panels do not feature a bezel and are mounted seamlessly together, a single continuous screen is created. The size of the screen is determined by the number of panels, while the pixel pitch determines the resolution. We realised the advantages of the LED technology, including the small size requirements and the ability to mount them directly onto a wall. However, due to the fragility of the LED panels, their use in a CAVE system was not feasible at that point.

### 3.1 Requirements

We decided that in order for us to consider LED panels for the construction of a CAVE, several key requirements had to be fulfilled:

1. Ability to resist smaller impacts without damage or scratches
2. Load-bearing capacity to allow multiple people to walk on them
3. Identical panels used for walls and floor for identical resolution and color
4. Integration of tracking cameras with the LED panels

In addition to the obvious fragility issue, we were also worried that the temperature and color shift of the panels would negatively impact a CAVE. LED panels heat up when in use, which would also heat up the interior space of a CAVE. Especially with a 5-sided CAVE, where a user is nearly surrounded by LED panels, the temperature could be too high for long-term use. Our LED powerwall also suffered from a color shift issue, where the colors would shift towards blue when viewed at a flat angle. In a CAVE this might lead to a color difference between walls due to different viewing angles and might negatively impact immersion.

<sup>1</sup>CAES's CAVE-like system

<sup>2</sup>HLRS's CAVE

<sup>3</sup>Villanova's CAVE

With the recent availability of LED panels that are robust enough to even walk on while still offering high resolution, we considered our requirements fulfilled. At this point, we considered LED to be the superior choice compared to projectors. One main expectation of the LED technology was to build a CAVE system with simpler construction and a smaller footprint than our projector-based installation.

We procured our system in late 2022, with construction of our LED CAVE starting in early 2023. The system has been in operation since summer 2023.

## 4 SPECIFICATIONS AND CONSTRUCTION

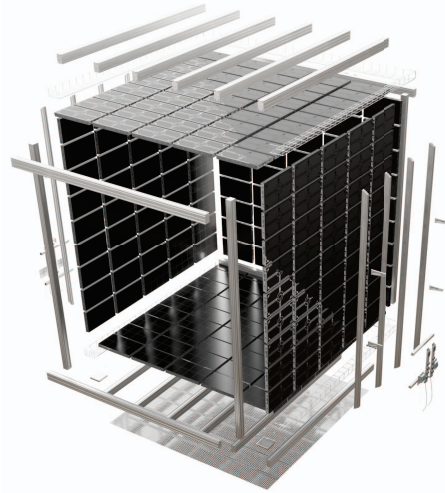


Figure 1: Structure and construction of the LED CAVE

Our CAVE is comprised of 5 borderless LED screens acting as left, right and back walls as well as ceiling and floor. The CAVE has a width and depth of 3 meters and a height of 3.04 meters. Each screen is constructed from 45 cabinets with each cabinet housing 8 LED panels (See Figure 1). The LED panels feature a pixel pitch of 1.25 mm resulting in a total resolution of  $2400\text{ pixels} \times 2430\text{ pixels}$  per wall. They provide a maximum brightness of up to 500 nits and an expected lifespan of up to 100,000 hours according to their datasheet. They are capable of displaying stereoscopic content with up to 120 frames per second (60 frames per second per eye) using shutter glasses.

For tracking, an ART tracking system with 4 ARTTRACK6/M cameras is used. The cameras are mounted in the corners near the ceiling. The cameras near the entrance are mounted outside the CAVE. For the cameras in the back, specialised LED panels were manufactured that provide a hole for a mounting strut to pass through the panel to hold the tracking camera.

The floor is able to withstand the weight of multiple people. While it is possible to walk directly on the LED panels, we employ the use of an acrylic protective sheet to prevent scratching of the panels by stones or abrasives from users' shoes over time. The sheet can be changed if necessary to maintain the quality of the floor.

One big advantage of LED technology is its space requirement compared to projector-based systems. The cabinets are mounted on a frame providing the basic structure. This results in a total size of the construction of about  $3.5\text{ m} \times 3.5\text{ m} \times 3.5\text{ m}$ , not counting additional hardware like the LED controller or cluster to operate the CAVE.

Table 1: Key data comparison of a projector-based CAVE and the LED CAVE.

|                     | Projector CAVE                                         | LED CAVE                                         |
|---------------------|--------------------------------------------------------|--------------------------------------------------|
| Scale               | $2.7\text{ m} \times 2.7\text{ m} \times 2.7\text{ m}$ | $3\text{ m} \times 3\text{ m} \times 3\text{ m}$ |
| Hardware per wall   | Two active stereo projectors                           | 45 cabinets with LED panels                      |
| Resolution per wall | $1920 \times 1920\text{ pixels}$                       | $2400 \times 2430\text{ pixels}$                 |
| Tracking system     | Optical tracking system (TRACKPACK/C)                  | Optical tracking system (ARTTRACK6/M)            |

#### 4.1 Comparison to Previous Projector-based CAVE

We compare the new LED CAVE to our previous, projector-based CAVE. Our previous system consisted of 5 acrylic screens being back-projected onto by 10 Christie Mirage WU3 projectors (two per wall). Tracking was also done with an ART tracking system. The cluster driving the installation was not upgraded and is identical for both the old and LED CAVE. Table 1 shows a comparison of the basic specifications of the old and new installations. In terms of construction, the biggest difference between the projector-based and LED CAVE is the space requirement. The projector-CAVE required projectors on the floors above and below the CAVE for the back-projected ceiling, respectively floor of the CAVE. Additionally, the projection was done via mirrors to fit the installation in the available space. In comparison, the new LED CAVE has a total size of about  $3.5\text{ m} \times 3.5\text{ m} \times 3.5\text{ m}$ , vastly reducing space requirements.

With the projector-based CAVE, alignment of the projectors proved to be a constant issue. For the images to be seamless and aligned properly, the projector images were recalibrated twice a year. Additionally, access to the projector area was strictly prohibited since any disturbance of the mirrors would cause the images of the projectors to go out of alignment. With the LED CAVE, we face no such issues. As the frame the cabinets are mounted on is a stable, static structure, there is no calibration or realignment required. This greatly reduces the required maintenance. In addition to the alignment, the maintenance also included changing the projectors' lamps due to their limited lifespan. This is another issue, that is no longer required for the LED CAVE, due to LEDs' expected lifespan of up to 100,000 hours. In summary, there is no regular maintenance work required on the LED CAVE. Another issue with the projectors was synchronisation. With both the old and LED CAVE being active stereo systems, the projectors needed to be synchronised to work with the shutter glasses. This was done by providing a synchronisation signal from the cluster rendering the images to all projectors. With the LED CAVE, a single LED controller (Novastar H15e) is driving the entire LED CAVE. The controller maps the input from the cluster nodes to the screens of the CAVE. As such, no additional synchronisation is required between the LED CAVE and the visualisation cluster, reducing the complexity of the system in comparison to the projector-based system.

## 5 FINDINGS AND DISCUSSION

In this section, we discuss our initial findings from operating the LED CAVE for several months as well as some early impressions from users and visitors. Comparisons between the old and new CAVE concerning optical quality are difficult since we could not operate both systems at once. In addition, we would expect the new system to outperform the old CAVE due to over ten years of technological development.

### 5.1 Early Impressions

Initial feedback to the new CAVE has been very positive, with users reporting favourably on the saturation of the colors, the brightness and an improved 3D effect. Users also report that the corners between the walls are less visible than our old CAVE when displayed



Figure 2: User in LED CAVE.

stereoscopic content. Depending on the application, some users report that corners of the LED CAVE are nearly imperceptible and that it is very hard to discern where the screens, and therefore the borders of the CAVE, are. Overall, the early feedback indicates that the LED CAVE provides a highly immersive experience. Specifically, user groups from the field of Art History reported that the quality of color representation is essential for their projects. They focus on representing their virtual models as true to reality as possible. Their feedback on the new LED CAVE is very positive, emphasizing the life-likeness of the virtual environments displayed in the CAVE.

Criticism of the new system was mainly concerned with the temperature within the CAVE. Due to the LED technology and the LED panels heating up during operation, temperatures within the CAVE are higher than with the old projector-based system. This will be discussed in detail in section Temperatures and Color Shift (5.3). Due to the minimal gap between the screens, air circulation is negligible and the temperatures within the CAVE tend to be higher than in the surrounding room.

### 5.2 Maintainability and Servicability

The LED CAVE is expected to not require regular maintenance. However, due to the large number of components like LED panels in use in the entire CAVE, smaller defects like dead pixels or occasional defective panels are to be expected.

As the LED panels can be serviced from the front, no access behind the CAVE is required and the cabinets may even be mounted onto a wall. We also maintain a stockpile of spare parts to be able to quickly repair any defects. Due to the robustness of the panels and the frame, most service work can be done by the team members. This includes swapping vital components like the LED panels or power supplies. For larger defects we do, however, rely on an external company for fault finding and repairs.

### 5.3 Temperatures and Color Shift

As predicted, there are two main issues with the new CAVE caused by the LED technology: higher temperatures in the CAVE as well as a color shift, depending on the viewing angle.

With the projector-based CAVE, all active components, most predominantly the projectors, were behind the acrylic screens. The



Figure 3: In these images a uni-colored grey image is shown in the LED CAVE to examine the color shift. Left: At this angle, the color shift of the LED CAVE is visible, with the right wall having a blue hue due to the flat angle. Right: A fish-eye view of the CAVE showing the color shift within the CAVE is not as prominent, with all walls having a similar color.

building is designed to completely separate the space in front and behind the CAVE with cooling being mainly for the area behind the CAVE. After installation of the LED CAVE this solution proved to be insufficient. While the cabinets also give off heat to the back, it is also radiated off the front as the panels heat up during operation, effectively heating up the inside of the CAVE and the surrounding room. Operating the LED CAVE without any changes to our cooling setup, we hit temperatures above  $30^{\circ}\text{C}$  in the surrounding room. To resolve this issue, we adapted the airflow, installing outlets in the room and using the cooling system for the projectors to cool the room in front of the CAVE. We measure the room temperature outside the CAVE, which is approx.  $20^{\circ}\text{C}$  without the CAVE on and about  $22^{\circ}\text{C}$  with the CAVE on (the peak temperature is at  $23^{\circ}\text{C}$ ). We collected data during a typical use case, with the temperature at the center of the CAVE peaking at about  $29^{\circ}\text{C}$ . Team members and users report that while the temperature is perceived as warm, it is still possible to use the CAVE regularly and for prolonged work.

The color shift is less obvious than anticipated. Most users do not report noticing any color shift at all (see Figure 3 Right). As expected, the color shift is dependent on two factors: the content displayed and the position of a user. Displaying uni-colored surfaces, especially when colored grey, there is a noticeable shift (See Figure 3). With colorful virtual environments and intense colours there is no noticeable color shift. The color shift is most noticeable from outside the CAVE, where users look at some of the screens at a very flat angle. When being inside the CAVE users would have to position themselves very close to a wall to achieve a similar effect.

As anticipated, there are some smaller defects, e.g. individual LEDs. So far, only a few users besides the team has addressed these, so we deem these as manageable.

#### 5.4 Limitations of this Paper

When discussing the advantages and challenges of the LED CAVE, we can only rely on our own experience. So far, feedback has been gathered verbally from user groups who had experience with our old CAVE. For future work, we aim to switch to a formal methodology with questionnaires. Additionally, the LED CAVE was officially opened in the summer of 2023, which limits our experience to only a few months.

Comparisons to other CAVE systems also prove difficult. A comparison to our old system provides limited insight since we could not operate both, the old and new CAVE, at the same time and, therefore were unable to conduct side-by-side comparisons. In addition, the old CAVE relied on technology that was 10 years old and outdated by the standards of today [18, 19]. Another issue is that we have not found an existing established scientific methodology to evaluate the added value of dedicated visualisation centers. However, with the many science domains that have profited from our support, we hope that they and research teams who are about to establish a

similar facility as well as the centers using other types of large-scale immersive displays can benefit from our report on this system.

#### 5.5 Future Work

Future work needs to be separated into two areas: collecting additional data on immersion / interaction in the LED CAVE and the possibilities opened up by the technology for VR installations. We aim to collect feedback from users about the visual quality and level of immersion provided by the LED CAVE. This can be done by providing questionnaires to users who either work regularly with or just visit the LED CAVE. In addition, we will collect data like occurring issues and failure rates to continue evaluating the technology for use in future projects. Further, we will examine interactions within the CAVE. We are already conducting research into using the LED CAVE as a virtual space connected to real and dynamic environments. This means that acquired sensor information from environments can be rendered into high-performance point clouds with distinctive objects.

LED technology opens up a variety of new possibilities for the future of CAVEs and CAVE-systems. With LED panels, it is possible to build screens of near arbitrary size and also non-rectangular shapes. Additionally, the frame and cabinets can be mounted very close to a wall, opening up the possibility of turning entire rooms into virtual environments. However, some issues like the heat could prove to be a limiting factor for such projects. However, we consider LED technology our choice for future projects like powerwalls or CAVEs.

#### 6 CONCLUSION

This paper reports on the early experiences with a CAVE based on LED technology for scientific visualisations and research. We examine the concept and implementation of the system and discuss early findings. Advantages include reduced maintenance, easier serviceability, favorable brightness, and improved color reproduction, among other things. Challenges include high temperatures, color shift and smaller defects like dead pixels. The LED CAVE system has gathered positive feedback from first impressions gathered from users. The next step is to thoroughly examine the advantages of such a system with long-term user studies.

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